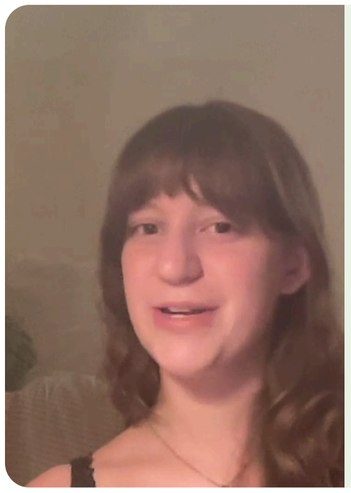
Meet your instructors: undergraduates teaching undergraduates.

UNDERGRADUATE INSTRUCTOR · LESSONS 1 & 2

Chloe King

Unix navigation and Bash scripting



FROM CHLOE'S OWN SESSION

1. Moves the fridge into the kitchen

```
HouseGame % mv Decoration/fridge.txt House/Kitchen/
```

2. Creates a new oven and describes it

```
Kitchen % touch oven.txt
Kitchen % nano oven.txt
Kitchen % cat oven.txt
A place to bake your cookies!
```

3. Reads the first wine orders

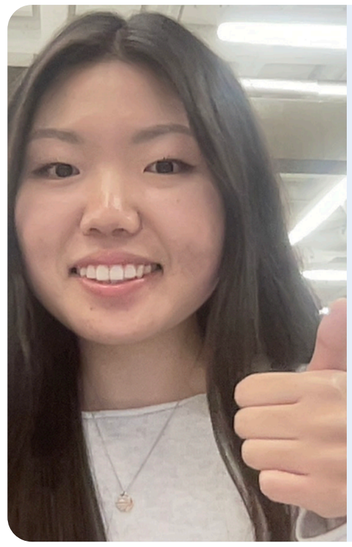
```
WineGame_New % cat orders_round1.csv
```

> "Even something as simple as having two spaces instead of one can completely make the code stop working."
>
> CHLOE, IN HER OWN WORDS

UNDERGRADUATE INSTRUCTOR · LESSON 3

Lindsey Ko

MRI file formats and conversion



FROM LINDSEY'S OWN SESSION

1. Looks inside a raw scan folder

```
sub-001 % cd FLAIR_DICOM
FLAIR_DICOM % ls
1.2.840.113619.6.495…-4-1-z43833.dcm    …
```

2. Converts it with dcm2niix

```
MRIcron % dcm2niix -f "sub-001_FLAIR_GUI" -p y -z y -o
…/gui_niftis
Found 44 DICOM file(s)
```

3. Checks what came out

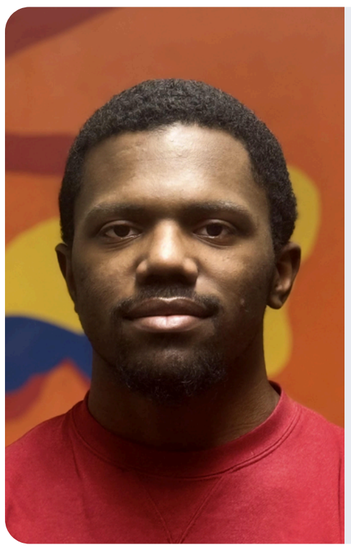
```
ScanConverterGame % ls gui_niftis
sub-001_FLAIR_GUI.json    sub-001_FLAIR_GUI.nii.gz
```

> "Once I had that installed and set up, the dcm2niix started working."
>
> LINDSEY, IN HER OWN WORDS

UNDERGRADUATE INSTRUCTOR · LESSON 4

Osiel Iglesias

Batch processing and remote computing



FROM OSIEL'S OWN SESSION

1. Looks at the whole dirty load

```
LaundryGame2 % cd dirty_laundry/round2_big_load
round2_big_load % ls
labcoat_dirty.png  pants_dirty.png  shirt_1_dirty.png  …
```

2. Logs in to the remote laundromat

```
LaundryGame2 % bash scripts/fake_ssh.sh
Connecting to ...
Password:
```

3. Checks the clean results

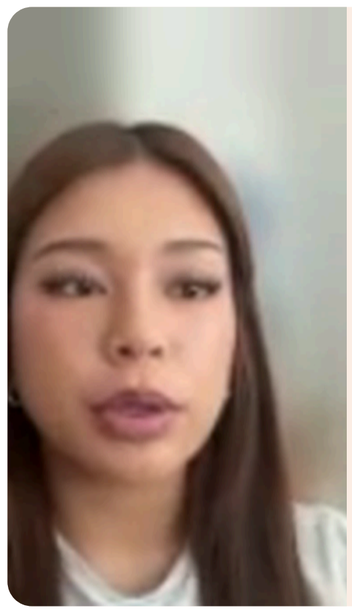
```
clean_laundry % ls
pants_clean.png  shirt_clean.png  socks_clean.png
```

> "…not to memorize every command, it's to understand the overall workflow."
>
> OSIEL, IN HIS OWN WORDS

UNDERGRADUATE INSTRUCTOR · LESSON 5

Tiffany Huang

Brain image review and lesion identification

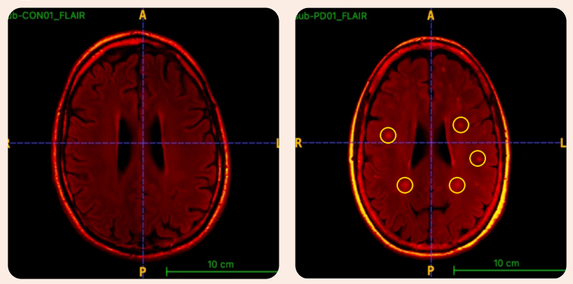


FROM TIFFANY'S OWN SESSION

1. Opens the lesion lab

```
~ % cd Desktop/ACLAB/LesionDetectiveGame_CAPSTONE
LesionDetectiveGame_CAPSTONE % ls
converted_niftis  lesion_masks  raw_dicoms  results  …
```

2. Compares two participants at a similar level



Control | Parkinson's disease

> "We combine all the skills that we've been learning into neuroimaging."
>
> TIFFANY, IN HER OWN WORDS